Phil McDonough
949-677-9450 | 1pjmcdonough@gmail.com | LinkedIn | Website

Highly adaptable and persistent problem solver with strong Python and R experience. Recent work focuses on AI development, including prompt engineering for large language models and training encoder-based ML models to extract structured insights from complex data.

Skills: Python, R
Certifications: PCEP—Certified Entry-Level Python Programmer, AWS Certified Cloud Practitioner

EXPERIENCE

Axon Remote
AI Research Intern June 2025–Present

- Executed applied AI research for Axon's beta-stage Escalation Detection product, advancing an early proof-of-concept (PoC) through model development, evaluation, and research on real-world constraints.
- Built the evaluation foundation for the Escalation Detection PoC by assembling a 187-transcript dataset, developing annotator guidelines, and creating a reusable evaluation workflow.
- Conducted LLM ablation studies across prompting, context windows, reasoning, and few-shot configurations, evaluating tradeoffs against product-relevant metrics.
- Trained and evaluated a BERT-based encoder model using synthetically generated data, demonstrating performance comparable to the LLM-based PoC.
- Evaluated voice activity detection models against body-camera power and bandwidth constraints; built a privacy-preserving PII redaction and repopulation workflow; and conducted counterfactual dialect evaluations to identify and mitigate responsible-AI risks.

PROJECTS

Fifth Third Bank Practicum Raleigh, NC
Tech Lead August 2024–April 2025

- Analyzed an unstructured text dataset with 7 million bankers notes using an LLM, Python, and Polars.
- Created features using NLP to cluster notes based on customer relationship health and product interest.
- Integrated features with pre-existing customer data to predict product adoption using boosting algorithms.

Elevance Health Capstone Blacksburg, VA
Team Member August 2022–December 2022

- Collaborated with the Individual Strategy and Analytics team to increase their understanding of market sizes in the US for the five main insurance types at the county level using Python for data collection and processing, and Tableau for dashboard visualizations.
- Executed a comprehensive 'Tools and Techniques' presentation, highlighting the array of aforementioned tools and techniques employed by our project team.

EDUCATION

MS in Analytics May 2025
Institute for Advanced Analytics, NC State University, Raleigh, NC

BS in Computational Modeling and Data Analytics, Economics Focus, Mathematics Minor May 2023
magna cum laude
Virginia Tech, Blacksburg, VA